

The GenAI Augmentation Paradox: Evaluating Synthetic Data in Low-Data Image Classification

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Abstract — We present a controlled study of image classification in a low-data regime on Tiny ImageNet. Our central question is whether synthetic images from pretrained diffusion models can recover performance lost when only 5% of training data is available, and whether any recovery reflects genuine representation improvement. We introduce three pipelines: Baseline (5% real), DiffusionBoost (5% real + synthetic), and Ceiling (100% real). Stage 1 establishes a data pipeline, trains a ResNet-18 baseline, and reports accuracy, calibration, robustness, and representation diagnostics. The model achieves 35.50% Top-1 accuracy with ECE 0.0372. Stage 2 will evaluate diffusion augmentation and class-dependent utility.

Keywords— Low-data classification, synthetic data augmentation, diffusion models, Tiny ImageNet, calibration (ECE), representation geometry, ResNet-18.

I. PROBLEM DEFINITION

We study image classification in a low-data regime on Tiny ImageNet (200 classes, 64×64 images). Our central question is whether synthetic images generated by pretrained diffusion models can recover performance lost when only 5% of the real training data is available, and whether any apparent recovery reflects genuine representation improvement rather than trivial data smoothing.

Formally, the input is an image $x \in \mathbb{R}^{3 \times 64 \times 64}$ and the output is a class label $y \in \{1, \dots, 200\}$. The low-data setting (25 images per class) introduces overfitting, miscalibration, and sensitivity to perturbations. We define three pipelines: (1) Baseline trained on 5% real data, (2) Ceiling trained on the full dataset, and (3) DiffusionBoost, introduced in Stage 2, which augments real data with class-conditional synthetic samples.

Stage 1 focuses on building a reliable data pipeline, training a pretrained ResNet-18, and establishing baseline metrics across accuracy, calibration, robustness, and representation geometry. We hypothesize that diffusion-generated images improve classification performance only when they introduce new semantic diversity, rather than merely smoothing the training distribution. This work provides a multi-axis evaluation framework for synthetic data in low-data regimes.

II. MINI LITERATURE SURVEY

Modern computer vision systems rely on large annotated datasets, yet many applications operate in data-scarce settings where models suffer from overfitting and unstable generalization. Pretrained models partially address this

limitation. [1] show that zero-shot models such as CLIP retain robustness under distribution shift, while naive fine-tuning degrades performance; they propose weight interpolation (WiSE-FT) to balance robustness and accuracy.

Diffusion models provide a generative approach to data augmentation. [2] introduce denoising diffusion probabilistic models, which learn the reverse of a Gaussian noising process. [3] extend this framework with latent diffusion, enabling high-resolution image synthesis efficiently in a compressed latent space.

Recent work demonstrates that diffusion-generated data can improve classification. [4] show that synthetic images improve ImageNet accuracy when combined with real data. [5] apply diffusion augmentation to long-tailed datasets, increasing intra-class diversity and improving classification performance. [6] propose diffusion-based augmentations that generate semantic variations, improving few-shot learning.

Combining synthetic and real data has also been explored. [7] introduce MixDiff, showing that synthetic data can improve representation learning and robustness. [8] demonstrate that augmentation strategies such as MixUp and CutMix improve robustness and generalization.

Representation-level analysis has received less attention in this context. [9] study representation similarity using CKA, enabling comparison of learned feature spaces across models. However, existing work largely focuses on accuracy gains, leaving open whether synthetic data improves representation quality or merely smooths decision boundaries.

Building on these studies, we evaluate diffusion-based augmentation in a controlled low-data setting using Tiny ImageNet [10]. We extend prior work by analyzing performance across accuracy, calibration [11], robustness, and representation geometry using a pretrained ResNet architecture [12]. Expected Calibration Error (ECE), introduced by [11], measures the discrepancy between predicted confidence and empirical accuracy.

III. EXPLORATORY DATA ANALYSIS

We use Tiny ImageNet as a controlled benchmark. The dataset contains 200 classes with 500 training and 50 validation images per class. To simulate a low-data regime, we construct a deterministic 5% stratified subset by sampling 25 images per class, yielding 5,000 training samples.

The subset is perfectly balanced, removing class imbalance as a confound. Figure 1 shows representative

samples across classes. All images are 64×64 pixels; a size histogram confirms uniform resolution. Channel statistics are mean = [0.483, 0.440, 0.384] and std = [0.265, 0.252, 0.269], consistent with ImageNet.



Fig. 1. Sample images from the 5% Tiny ImageNet subset across multiple classes

We compute the average inter-class cosine distance between pretrained feature embeddings (0.145), indicating moderate class separability despite limited data. Inputs are resized to 224×224 for compatibility with pretrained ResNet features. Standard augmentations (random crops, horizontal flips) are applied.

IV. BASELINE AND PROOF OF CONCEPT

Our baseline model is a ResNet-18 pretrained on ImageNet. The final layer is replaced with a 200-class classifier and fine-tuned on the 5% subset using cross-entropy loss and AdamW (learning rate 3×10^{-4} , batch size 64) for 5 epochs.

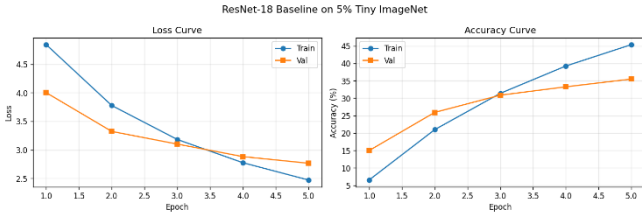


Fig. 2. Training Loss and validation accuracy over 5 epochs for the ResNet-18 baseline

The model achieves 35.50% Top-1 accuracy on the validation set, establishing a non-trivial baseline. Figure 2 shows training loss and validation accuracy.

We evaluate calibration using Expected Calibration Error (ECE = 0.0372), which measures confidence–accuracy mismatch. The reliability diagram (Figure 3) indicates mild overconfidence.

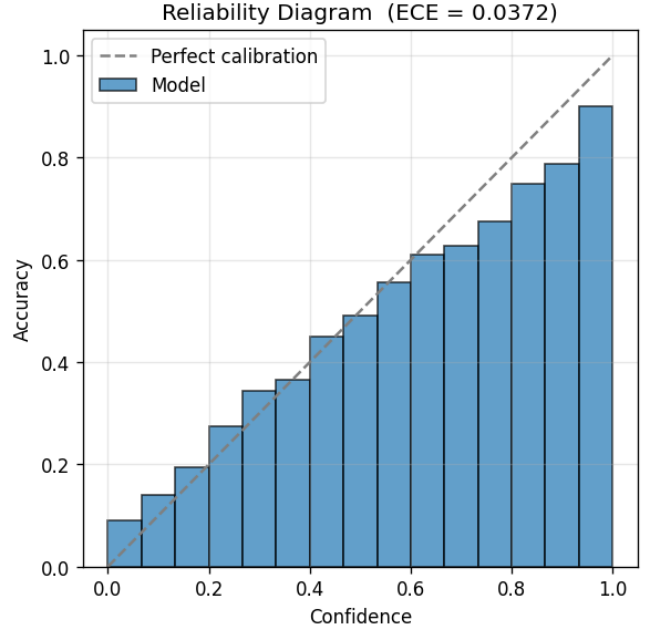


Fig. 3. Reliability diagram (ECE = 0.0372) on the full validation set

Robustness is evaluated under controlled corruptions: Gaussian noise ($\sigma = 0.1$), Gaussian blur (kernel size 5), and brightness shift (+0.2). Accuracy drops from 35.5% to 10.0%, 36.3%, and 31.8%, respectively, reflecting sensitivity to noise and moderate robustness to blur and brightness shifts.

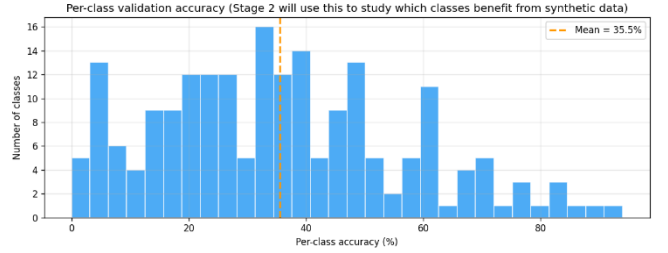


Fig. 4. Histogram of per-class validation accuracy (baseline)

Per-class accuracy varies substantially (mean 35.5%, range 0–94%, std 21.1%), providing a baseline for analyzing class-dependent effects of synthetic augmentation. Finally, penultimate-layer features are used for a linear probe, achieving 42.15% accuracy, and the covariance eigenvalue spectra are analyzed to characterize feature space structure.

In Stage 2, we will introduce diffusion-generated data and evaluate its effect across calibration, robustness, and representation structure, including CKA similarity and feature spectra, as well as class-dependent utility.

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